Understanding the ROMS bulk fluxes

From winds and humidity to stress and ocean heating

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The routine turns bulk atmospheric and ocean conditions into surface boundary fluxes.

Wind relative to the ocean determines the direction of momentum exchange. Temperature and humidity contrasts drive sensible and latent heat exchange. Atmospheric stability and surface roughness determine the transfer efficiency. These quantities depend on one another, so the routine updates them iteratively.

Scope. The main derivation follows the open-water, non-RegCM path in the inspected file, including optional cool-skin and wave-roughness branches. Sea-ice extensions are identified separately. This is an explanation of the inspected source, not confirmation of the flags enabled in any particular executable. No model code was modified.

A consistent notation

Handout symbol	Code name	Meaning
$U_{\text{wind}}, V_{\text{wind}}$	`Uwind,Vwind`	Earth-relative wind components
$u_{\text{oce}}, v_{\text{oce}}$	centered surface `u,v`	Ocean surface-current components
$U_{\text{rel}}, V_{\text{rel}}$	`Uair,Vair`	Relative wind components
$U_{\text{rel,mag}}$	`Wmag`	Relative vector-wind magnitude
u_{gust}	`Wgus`	Convective gustiness speed
U_{eff}	`delW,Wspeed`	Gustiness-enhanced exchange speed
u_* (blue)	`Wstar`	Friction velocity (m/s)
T_* (orange)	`Tstar`	Temperature scale (K)
q_* (green)	`Qstar`	Specific-humidity scale (kg/kg)

Wstar is friction velocity, not gustiness.

Stars are subscripts throughout. In particular, u_{gust} is not u_*, and U_{eff} is not the magnitude used to normalize the final stress direction.

Atmospheric inputs and optional coupled fields

The standard forcing provides wind, temperature, pressure and humidity, together with the precipitation and radiation fields appropriate to the selected options. Not every field in the table is required in every configuration. ROMS supplies the ocean surface state. File units and internal units differ for humidity and radiation because the input processing applies conversions before the bulk-flux routine is called.

Forcing variable	Quantity	Forcing-file units
`Uwind`	Earth-relative wind, first horizontal component	m s^{-1}
`Vwind`	Earth-relative wind, second horizontal component	m s^{-1}
`Tair`	Near-surface air temperature	°C
`Pair`	Surface air pressure	millibar (hPa)
`Qair`	Relative humidity (default input configuration)	percent
`rain`	Precipitation mass flux	kg m^{-2} s^{-1}
`swrad`	Net downward shortwave radiation	W m^{-2}
`lwrad_down`	Optional downward longwave; define `LONGWAVE_OUT`	W m^{-2}
`lwrad`	Net downward longwave, as an alternative	W m^{-2}
`cloud`	Cloud fraction, if empirical longwave is selected	fraction, 0–1
`PBLh`	Optional WRF boundary-layer depth; `PBLH` (experimental)	m

Relative humidity is a fraction inside the calculation.

The inspected default variable metadata reads `Qair` in percent and multiplies it by 0.01. Thus 80% becomes RH = 0.8. An alternative specific-humidity input requires consistent metadata and units of g kg^{-1}. The internal threshold described later acts on the converted input, not directly on percentage values in the forcing file. The external forcing name `Qair` should be distinguished from the routine's saturation-humidity intermediate.

Radiation uses a positive-downward convention.

Radiative forcing is supplied in W m^{-2}. Input processing converts it to kinematic temperature-flux units; the routine recovers W m^{-2} using $\rho_0 c_{p,o}$ for its internal heat budget. When downward longwave is available, define `LONGWAVE_OUT` and supply `lwrad_down`. Otherwise select the prescribed-net or empirical longwave option with its corresponding inputs. These are alternative radiation pathways.

Ocean state and measurement heights complete the inputs.

The upper ocean temperature (°C), surface current components (m s^{-1}), and surface density are taken from ROMS. Winds and currents must share coordinate axes before subtraction. Wind, temperature and humidity measurement heights are in metres (10 m defaults). Wave-dependent roughness additionally uses significant wave height (m), peak period (s), and wavelength (m), as required by the selected option. For WRF coupling, `PBLH` is an experimental option; this source requires both `FRC_COUPLING` and `PBLH` to accept the coupled field. Otherwise the parameterized depth is 600 m by default.

1. Wind minus currents: direction and exchange speed

With `WIND_MINUS_CURRENT`, the code first averages staggered surface ocean velocities to rho points and subtracts them from the wind as in Eq. (1):

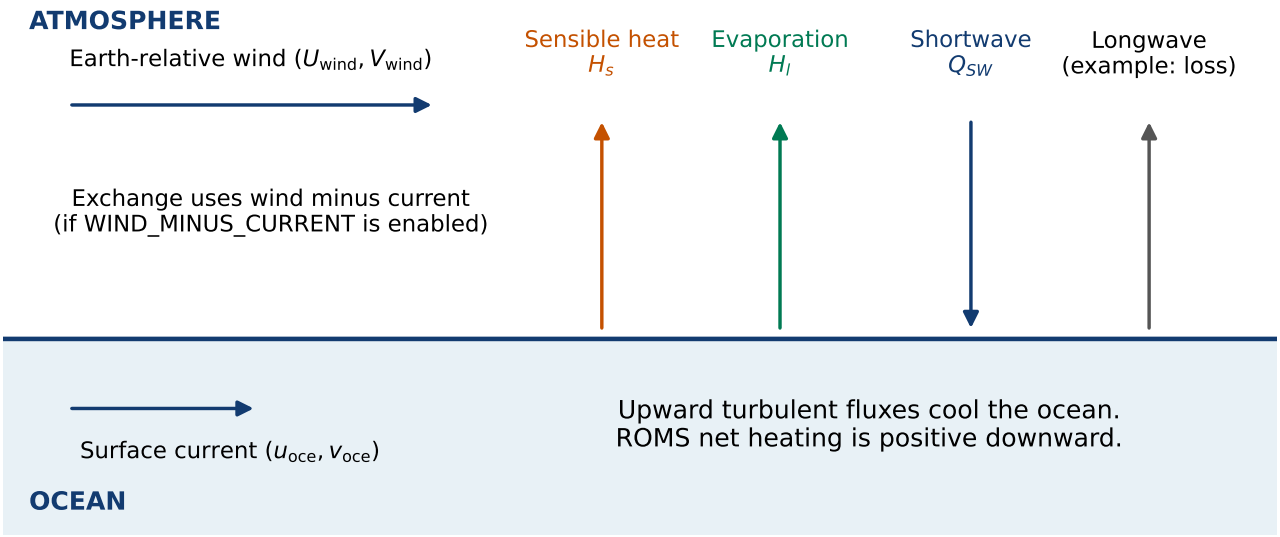
```
Uair(i,j)=Uwind(i,j)                                    &
 & -0.5_r8*(u(i,j,N(ng),nrhs)+u(i+1,j,N(ng),nrhs))
Vair(i,j)=Vwind(i,j)                                    &
 & -0.5_r8*(v(i,j,N(ng),nrhs)+v(i,j+1,N(ng),nrhs))
Wmag(i)=SQRT(Uair(i,j)**2+Vair(i,j)**2)
delW(i)=SQRT(Wmag(i)**2+Wgus(i)**2)
```

$$\boxed{U_{\rm rel} = U_{\rm wind} - u_{\rm oce}, \qquad V_{\rm rel} = V_{\rm wind} - v_{\rm oce}} \tag{1}$$

$$\boxed{U_{\rm rel,mag} = \sqrt{U_{\rm rel}^2 + V_{\rm rel}^2}, \qquad U_{\rm eff} = \sqrt{U_{\rm rel,mag}^2 + u_{\rm gust}^2}.} \tag{2}$$

Without that flag, $U_{\rm rel} = U_{\rm wind}$ and $V_{\rm rel} = V_{\rm wind}$: no current is subtracted. Special eastern/northern edge assignments replace the interior averaging at those boundary points.

Coordinate convention. "Earth-relative" means the wind is measured relative to the Earth rather than the moving ocean. All subtracted components must nevertheless be expressed in the same axes. On a rotated grid, the source expects model-axis components; applicable input paths in `set_data.F` rotate winds before this routine. One must not subtract east/north winds directly from unrotated grid-axis currents.



Exchange schematic. The upward arrows illustrate an evaporating, ocean-cooling state, not a required sign for every flux. Gustiness enhances turbulent exchange but is not a directional vector added to the resolved wind.

Numerical protection. The numerical constant is $\boxed{\varepsilon = 10^{-20}}$. We retain ε wherever the source adds it or uses it as a floor. It is a tiny numerical safeguard, not the water-vapor/dry-air molecular-mass ratio. Not every denominator in this source is protected.

2. Humidity: from vapor pressure to mass fraction

Specific humidity is a mass fraction; relative humidity is a saturation ratio. They are not interchangeable.

Let $q = m_v/(m_d + m_v)$ be specific humidity and $r = m_v/m_d$ the mixing ratio. If e is water-vapor partial pressure, p total pressure, and $M_v/M_d = 0.62197$, the ideal-gas mixture relations give

$$r = 0.62197\frac{e}{p-e}, \qquad q = \frac{r}{1+r} = \frac{0.62197e}{p-(1-0.62197)e}. \tag{3}$$

The code uses 0.378 for $1-0.62197$. It evaluates saturation vapor pressure in hPa (mbar) using Celsius temperature T_C:

$$\boxed{e_s(T_C, p) = (1.0007 + 3.46\times10^{-6}p)\,6.1121\exp\left(\frac{17.502T_C}{240.97+T_C}\right).} \tag{4}$$

```
cff=(1.0007_r8+3.46E-6_r8*PairM)*6.1121_r8*             &
 & EXP(17.502_r8*TairC(i)/(240.97_r8+TairC(i)))
Qair(i)=0.62197_r8*cff/(PairM-0.378_r8*cff+eps)
IF (RH.lt.2.0_r8) THEN
  cff=cff*RH
  Q(i)=0.62197_r8*cff/(PairM-0.378_r8*cff+eps)
ELSE
  Q(i)=RH/1000.0_r8
END IF
```

For fractional relative humidity RH, the routine uses Eq. (5):

$$e_a = \mathrm{RH}e_s(T_a, p), \quad q_{a,\mathrm{sat}} = \frac{0.62197e_s(T_a,p)}{p-0.378e_s(T_a,p)+\varepsilon}, \quad \boxed{q_a = \frac{0.62197e_a}{p-0.378e_a+\varepsilon}.} \tag{5}$$

Saturated and actual humidity: `Qair` is $q_{a,\mathrm{sat}}$, while `Q` is the actual q_a. Applying relative humidity to vapor pressure before converting to q preserves the nonlinear denominator. In general, $q_a \neq \mathrm{RH}q_{a,\mathrm{sat}}$ exactly.

Example, not model output. At 25°C, $p = 1013$ hPa and $\mathrm{RH} = 0.8$, this formula gives $e_s = 31.804$ hPa, $q_{a,\mathrm{sat}} = 19.762$ g/kg and $q_a = 15.771$ g/kg. Multiplying saturation specific humidity by 0.8 would give 15.809 g/kg instead.

Input caveat. The branch interprets `Hair`<2 as fractional RH, otherwise as specific humidity in g/kg. It does *not* interpret an input value of 80 as 80% RH here. The heuristic also cannot distinguish very dry specific-humidity input below 2 g/kg from fractional RH; forcing units must be checked upstream.

3. Sea-surface humidity and thermodynamic contrasts

Seawater lowers saturation vapor pressure, and warmer water usually raises it strongly.

The surface saturation pressure is reduced by a fixed salinity factor:

$$e_{s,\mathrm{sea}} = 0.98\, e_s(T_s, p), \qquad \boxed{q_s = \frac{0.62197 e_{s,\mathrm{sea}}}{p - 0.378 e_{s,\mathrm{sea}}}.} \tag{6}$$

There is **no added** ε in this particular source denominator. The factor 0.98 approximates reduced water activity over seawater; it does not use the model salinity value. The strong temperature dependence of saturation vapor pressure reflects the thermodynamics of phase equilibrium (Clausius–Clapeyron behavior); the implemented exponential is an empirical approximation, not a direct integration of that law.

Let T_s be the upper ocean temperature and T_a the air temperature. Define the contrasts used in the similarity scales as

$$\Delta T = T_s - T_a - \delta T_{\mathrm{skin}}, \qquad \Delta q = q_s - q_a - \delta q_{\mathrm{skin}}. \tag{7}$$

The lowercase quantities δT_{skin} and δq_{skin} are the skin cooling and its associated saturation-humidity reduction. Both are zero unless `COOL_SKIN` is defined. Positive skin cooling lowers the interface temperature and humidity relative to the bulk ocean values. Rain heat exchange uses the uncorrected bulk differences, stated explicitly below.

Moist-air density and material properties

The file converts Celsius to Kelvin using $T_K = T_C + 273.16$, and hPa to Pa using $100p$:

$$\boxed{\rho_a = \frac{100p}{R_d T_{aK}(1 + 0.61 q_a)}}, \qquad R_d = 287.1\ \mathrm{J\,kg^{-1}K^{-1}}. \tag{8}$$

The factor $1 + 0.61 q_a$ is the virtual-temperature correction: moist air is less dense than dry air at the same temperature and pressure. This is why humidity affects buoyancy as well as evaporation.

$$\nu_a = 1.326 \times 10^{-5}\left[1 + T_{aC}\left(6.542 \times 10^{-3} + T_{aC}(8.301 \times 10^{-6} - 4.84 \times 10^{-9} T_{aC})\right)\right], \tag{9}$$

$$L_v = (2.501 - 0.00237 T_{sC}) \times 10^6\ \mathrm{J/kg}. \tag{10}$$

The air viscosity ν_a is in $\mathrm{m^2/s}$. L_v decreases with temperature. Under `REGCM_COUPLING`, the source instead fixes $L_v = 2.5104 \times 10^6$ J/kg.

4. What the three similarity scales mean

The scales summarize momentum, heat and moisture exchange in the atmospheric surface layer.

Ignoring numerical floors for this physical interpretation, the turbulent flux relations are

$$\boxed{\tau_{\text{turb}} = \rho_a {u_*}^2}, \qquad \boxed{H_s = -\rho_a c_{pa} u_* T_*}, \qquad \boxed{H_l = -\rho_a L_v u_* q_*}. \tag{11}$$

Here H_s and H_l are positive upward, out of the ocean. A warm, evaporating surface normally gives $T_* < 0$ and $q_* < 0$, so the minus signs yield positive ocean heat loss. Condensation or heat transfer into the ocean can reverse the corresponding sign. u_* is a nonnegative velocity scale, not the mean ocean or wind speed.

For observation heights z_u, z_T, z_q, Eqs. (12)–(14) give the full similarity relations:

$$u_* = \max\left(\varepsilon, \frac{\kappa U_{\text{eff}}}{\ln\left(\frac{z_u}{z_{0m}}\right) - \psi_u\left(\frac{z_u}{L}\right)}\right), \tag{12}$$

$$T_* = -\frac{\kappa \Delta T}{\ln\left(\frac{z_T}{z_{0T}}\right) - \psi_h\left(\frac{z_T}{L}\right)}, \tag{13}$$

$$q_* = -\frac{\kappa \Delta q}{\ln\left(\frac{z_q}{z_{0q}}\right) - \psi_h\left(\frac{z_q}{L}\right)}. \tag{14}$$

Here ΔT and Δq already include the optional skin reductions. The momentum stability function ψ_u is the same function denoted ψ_m elsewhere in this handout. The source floors friction velocity after division; the logarithmic denominators have no added numerical constant. The von Karman constant is $\kappa = 0.41$.

Why iteration? Roughness depends on u_*; stability depends on u_*, T_*, q_*; these scales depend on roughness and stability. Gustiness depends on buoyancy flux, which again depends on the scales. The algorithm closes this loop rather than prescribing a single constant drag coefficient.

5. Initialization before the three-pass loop

The initial guesses are more than arbitrary starting values. They use neutral transfer estimates and a bulk Richardson number to enter the nonlinear loop near a plausible solution.

$$u_{\mathrm{gust}}^{(0)} = 0.5\ \mathrm{m/s}, \qquad z_{0m}^{\mathrm{seed}} = 10^{-4}\ \mathrm{m}, \qquad U_{10} = U_{\mathrm{eff}}\frac{\ln(10/z_{0m}^{\mathrm{seed}})}{\ln(z_u/z_{0m}^{\mathrm{seed}})}, \tag{15}$$

$$u_*{}^{\mathrm{seed}} = 0.035U_{10}, \quad z_{0,10} = 0.011\frac{(u_*{}^{\mathrm{seed}})^2}{g} + \frac{0.11\nu_a}{u_*{}^{\mathrm{seed}}}, \quad C_{D10} = \left[\frac{\kappa}{\ln(10/z_{0,10})}\right]^2. \tag{16}$$

The initialization uses $C_{H10} = 0.00115$, $C_{t10} = C_{H10}/\sqrt{C_{D10}}$, and

$$z_{0T,10} = 10\exp(-\kappa/C_{t10}), \quad C_D = \left[\frac{\kappa}{\ln(z_u/z_{0,10})}\right]^2, \quad C_t = \frac{\kappa}{\ln(z_T/z_{0T,10})}, \quad C_C = \frac{\kappa C_t}{C_D}. \tag{17}$$

The initial stability estimate is

$$Ri_b = -\frac{gz_u[\Delta T + 0.61T_{aK}(q_s - q_a)]}{T_{aK}U_{\mathrm{eff}}^2 + \varepsilon}, \qquad Ri_{bc} = -\frac{z_u}{0.004z_i\beta_g^3}, \tag{18}$$

$$\zeta^{(0)} = \begin{cases} C_C Ri_b/(1 + Ri_b/Ri_{bc}), & Ri_b < 0, \\ C_C Ri_b/(1 + 3Ri_b/C_C), & Ri_b \geq 0, \end{cases} \qquad L^{(0)} = z_u/\zeta^{(0)}. \tag{19}$$

z_i is the atmospheric boundary-layer depth: 600 m by default, or the coupled PBL height when enabled. $\beta_g = 1.2$. Initial cool-skin temperature reduction is 0.3 K when enabled, otherwise zero; the initial humidity correction is zero.

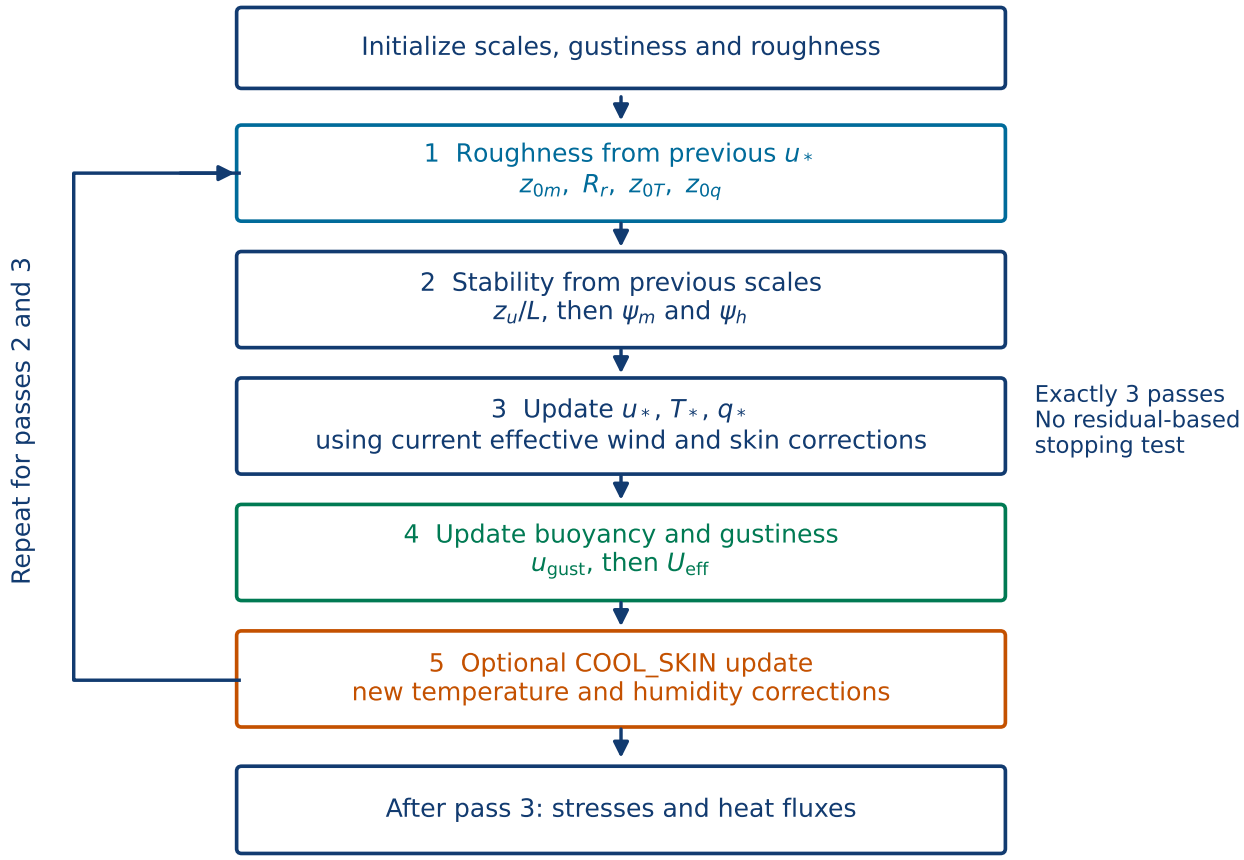
The source then computes first guesses of all three scales using the similarity equations in Section 4 with $L^{(0)}$, $z_{0,10}$ and $z_{0T,10}$. It sets the Charnock coefficient before the loop (Section 7).

Numerical fidelity. The initial viscous roughness term divides by u_*^{seed} without ε, and $L^{(0)} = z_u/\zeta^{(0)}$ also has no ε. An exactly neutral initial state therefore takes a different numerical path from the protected L update inside the loop. These equations describe the implementation; they are not a claim that every neutral or low-wind edge case has been safeguarded.

6. The actual iteration loop

There are exactly three passes, not an adaptive "iterate until converged" test.

```
integer, parameter :: IterMax = 3
DO Iter=1,IterMax
  ! roughness -> stability -> similarity scales
  ! -> gustiness -> optional cool-skin update
END DO
```



Order matters. Each pass uses the preceding state to update roughness and stability, then updates the scales, then gustiness and optional skin corrections. Updated gustiness/skin values feed the next pass.

After the third pass, the routine proceeds directly to fluxes. The final gustiness and skin updates are therefore not followed by an additional scale solve. The final stress and turbulent heat fluxes use the stored third-pass scales. A source comment says convergence usually occurs within three iterations, but there is no computed convergence residual or stopping tolerance.

7. Roughness controls transfer efficiency

Momentum roughness combines a wave-related term and a viscous term.

For the default branch without a wave-specific roughness option, Eq. (20) gives

$$\boxed{z_{0m} = \alpha_{\mathrm{Ch}} \frac{u_*^{\,2}}{g} + \frac{0.11\nu_a}{u_* + \varepsilon}}, \qquad R_r = \frac{z_{0m} u_*}{\nu_a}. \tag{20}$$

The COARE 3.5 branch sets, before the loop,

$$\alpha_{\mathrm{Ch}} = \min(0.028, -0.005 + 0.0017 U_{\mathrm{eff}}). \tag{21}$$

This exact expression has no lower clipping in the file. At low speed it can be negative; the viscous term remains in the roughness expression. The handout does not replace it silently with a different COARE formula. `DRAGLIM_DAVIS`, if enabled, subsequently bounds z_{0m} between 1.27×10^{-7} and 2.85×10^{-3} m.

For `COARE_30`, $\alpha_{\mathrm{Ch}} = 0.011$ below 10 m/s, increases linearly to 0.018 at 18 m/s, and stays at 0.018 above that speed. These wind thresholds use the initial gustiness-enhanced speed in this routine.

Thermal and moisture roughness:

$$\boxed{z_{0q} = z_{0T} = \min\left(1.6 \times 10^{-4}, \frac{5.8 \times 10^{-5}}{R_r^{0.72}}\right)} \quad \text{(COARE 3.5)}, \tag{22}$$

$$z_{0q} = z_{0T} = \min\left(1.15 \times 10^{-4}, \frac{5.5 \times 10^{-5}}{R_r^{0.6}}\right) \quad \text{(COARE 3.0)}. \tag{23}$$

These are empirical lengths in metres. Distinct momentum and scalar roughness lengths represent the different molecular and turbulent transfer constraints near the interface.

Optional wave-specific replacements for the first term of z_{0m}:

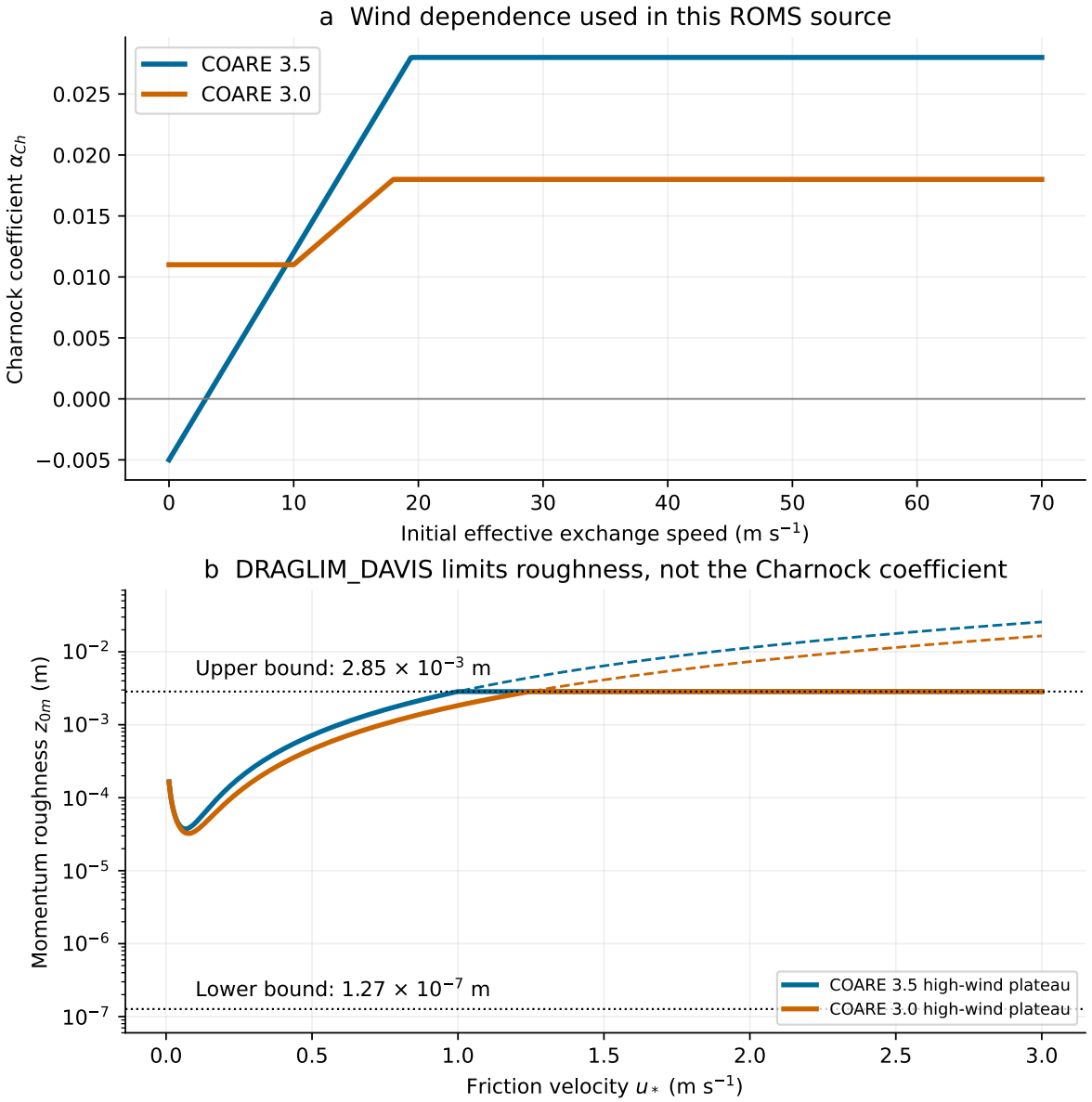
$$\text{Oost:} \quad \frac{25}{\pi} \lambda_p (u_*/c_p)^{4.5}, \tag{24}$$

$$\text{Taylor–Yelland:} \quad 1200 H_{\mathrm{wave}} (H_{\mathrm{wave}}/\lambda_p)^{4.5}, \tag{25}$$

$$\text{Drennan:} \quad 3.35 H_{\mathrm{wave}} [\min(u_*/c_p, 0.1)]^{3.4}. \tag{26}$$

Each retains $0.11\nu_a/(u_* + \varepsilon)$. Here H_{wave} is significant wave height (code `Hwave`), distinct from sensible heat flux H_s. Wave phase speed is $c_p = \lambda_p / \max(T_p, \varepsilon)$, or $g \max(T_p, \varepsilon)/(2\pi)$ for the deep-water option. These branches are alternatives with preprocessor precedence, not additive corrections.

Charnock coefficient and high-wind drag limits



Exact coefficient curves; conditional roughness curves. Panel (a) uses the initial effective exchange speed in the inspected source, including its negative COARE 3.5 coefficient below about 2.94 m/s. Panel (b) holds each coefficient at its high-wind plateau and varies friction velocity, with air viscosity 1.5×10^{-5} m^2/s. Dashed curves are unlimited; solid curves include `DRAGLIM_DAVIS`. These are parameterization diagnostics, not simulated hurricane winds.

Recommendation for hurricane applications.

Enable `DRAGLIM_DAVIS` as a practical high-wind option and assess sensitivity against observations. It bounds z_{0m} between 1.27×10^{-7} and 2.85×10^{-3} m. Under neutral conditions at 10 m, these correspond to $C_D = [\kappa/\ln(10/z_{0m})]^2$ of approximately 0.000508 and 0.00252. Stability changes the actual transfer coefficient. This limiter prevents continued roughness growth but does not by itself establish hurricane flux accuracy or account for all sea-state and spray effects.

8. Stability and gustiness close the loop

Let $\zeta = z_u/L$. The code calculates

$$\boxed{\zeta = \frac{\kappa g z_u [T_*(1 + 0.61 q_a) + 0.61 T_{aK} q_*]}{T_{aK} {u_*}^2 (1 + 0.61 q_a) + \varepsilon}}, \qquad L = \frac{z_u}{\zeta + \varepsilon}. \tag{27}$$

It evaluates $\psi_m(\zeta)$ for momentum and $\psi_h(z_T/L)$, $\psi_h(z_q/L)$ for heat and moisture. Negative ζ denotes unstable stratification, typically associated with upward buoyancy transfer. Positive ζ denotes stable stratification. The exact piecewise stability functions appear in Appendix A.

Gustiness allows scalar exchange even when the resolved mean wind is weak.

The implemented buoyancy quantity is, literally,

```
Bf=-g/(TairK(i)+0.61_r8*Qstar(i))*Wstar(i)*              &
 & (Tstar(i)+0.61_r8*TairK(i)*Qstar(i))
IF (Bf.gt.0.0_r8) THEN
  Wgus(i)=blk_beta*(Bf*blk_Zabl)**(1.0_r8/3.0_r8)
ELSE
  Wgus(i)=0.2_r8
END IF
delW(i)=SQRT(Wmag(i)**2+Wgus(i)**2)
```

$$B_{\text{code}} = -\frac{g u_* (T_* + 0.61 T_{aK} q_*)}{T_{aK} + 0.61 q_*}, \qquad \boxed{u_{\text{gust}} = \begin{cases} \beta_g (B_{\text{code}} z_i)^{1/3}, & B_{\text{code}} > 0, \\ 0.2 \text{ m/s}, & B_{\text{code}} \le 0, \end{cases}} \tag{28}$$

$$U_{\text{eff}} = \sqrt{U_{\text{rel,mag}}^2 + u_{\text{gust}}^2}. \tag{29}$$

The coupled-PBL option replaces `blk_Zabl` by the supplied boundary-layer depth. The positive-buoyancy branch does not separately impose a 0.2 m/s minimum.

Source-fidelity note. The denominator $T_{aK} + 0.61 q_*$ is exactly what the inspected source contains. It is not the usual virtual-temperature form $T_{aK}(1 + 0.61 q_a)$ and is dimensionally unusual as written. It should be checked against the intended reference implementation before interpreting it as a first-principles thermodynamic identity. We have not changed it. Similarly, this expression contains no added ε in its denominator.

9. Optional cool-skin correction

The bulk ocean temperature need not equal the temperature at the molecular interface.

Under `COOL_SKIN`, an interface cooling estimate modifies both thermal and humidity contrasts. The humidity slope used in the source is

$$C_{\text{wet}} = \frac{0.622 L_v q_s}{R_d T_{sK}^2}, \qquad \delta q_{\text{skin}} = C_{\text{wet}} \delta T_{\text{skin}}. \tag{30}$$

This is a linearized saturation-humidity response to skin cooling. It links latent heat transfer to the interface thermodynamics rather than treating humidity independently of skin temperature.

Each pass starts its skin-thickness calculation from $h_c = 0.001$ m and computes

$$f_c = 0.065 + 11 h_c - \frac{(1 - e^{-1250 h_c}) 6.6 \times 10^{-5}}{h_c}, \tag{31}$$

$$H_{sb} = -\rho_a c_{pa} u_* T_*, \qquad H_{lb} = -\rho_a L_v u_* q_*, \tag{32}$$

$$\boxed{Q_{\text{cool,code}} = Q_{LW} + H_{sb} + H_{lb} - f_c Q_{SW}}, \qquad Q_b = \alpha_w Q_{\text{cool,code}} + \beta_w H_{lb} \frac{c_{pw}}{L_v}. \tag{33}$$

Here α_w and β_w are the water thermal expansion and saline contraction coefficients passed into this routine. They are unrelated to the wave-force alpha multipliers in the earlier handout.

$$C_\lambda = \frac{16 g c_{pw} (\rho_w \nu_w)^3}{k_w^2 \rho_a^2}. \tag{34}$$

When both $Q_{\text{cool,code}} > 0$ and $Q_b > 0$,

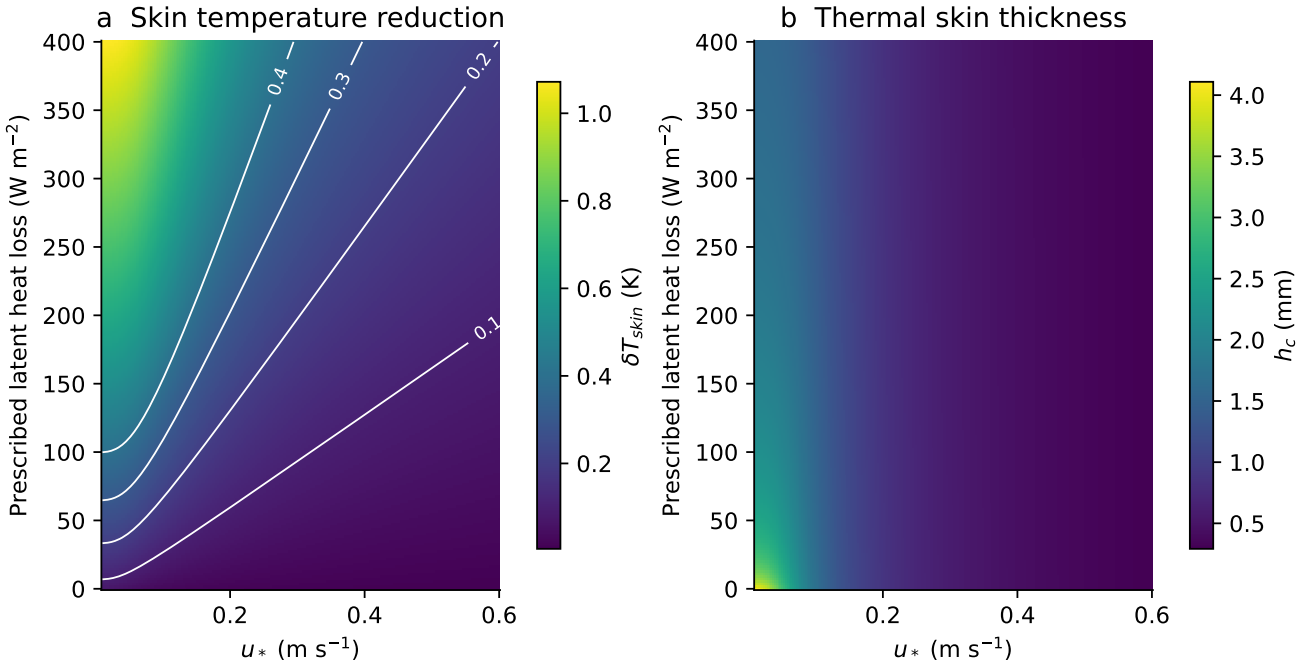
$$\lambda_c = \frac{6}{\left[1 + (C_\lambda Q_b / (u_* + \varepsilon)^4)^{0.75}\right]^{1/3}}, \qquad h_c = \frac{\lambda_c \nu_w}{\sqrt{\rho_a / \rho_w} u_* + \varepsilon}, \tag{35}$$

$$\delta T_{\text{skin}} = \frac{Q_{\text{cool,code}} h_c}{k_w}. \tag{36}$$

Otherwise $\delta T_{\text{skin}} = 0$; then δq_{skin} is updated. Defaults are $\nu_w = 10^{-6}$ m^2/s, $k_w = 0.6$ W/(m K), and $c_{pw} = 4000$ J/(kg K).

Sign audit, not a silent correction. The file adds `LRad` in `Qcool`, although the longwave calculation and final heat budget use positive-downward net longwave. The expression above reproduces that code. Its longwave sign deserves a targeted check if this option is used; do not read it as an independently validated outward-cooling balance. This file does not implement a separate prognostic diurnal warm-layer calculation in the described open-water path.

When does cool skin matter?



Weak mechanical mixing can allow a larger temperature drop across the skin.

The panels evaluate the skin submodel with prescribed latent heat loss and u_*. Higher evaporation supplies stronger cooling; increasing u_* thins the thermal layer. Buoyancy-driven renewal also limits its thickness, so the temperature reduction does not grow without bound as mechanical stress decreases.

Controlled sensitivity calculation. Shortwave and net longwave are set to zero to isolate turbulent cooling and avoid the longwave sign issue identified in Section 9. Sensible heat loss is 10 W/m^2; latent loss varies from 0 to 400 W/m^2. Fixed properties are $\rho_a = 1.2$, $\rho_w = 1025$ kg/m^3, $\nu_w = 10^{-6}$ m^2/s, $k_w = 0.6$ W/(m K), $c_{pw} = 4000$ J/(kg K), $L_v = 2.43\times10^6$ J/kg, $\alpha_w = 2.5\times10^{-4}$ K^{-1} and $\beta_w = 0.026$. These are illustrative prescribed values. The calculation is not the full iterative atmospheric bulk-flux solution; not all combinations are necessarily sustained by a given atmosphere.

Low vector wind is not zero exchange. Convective gustiness can maintain U_{eff} and evaporation even when the resolved relative wind is small. Strong evaporation additionally requires a humidity deficit and sufficient exchange. Therefore the horizontal axis uses friction velocity rather than assigning a unique vector wind to each heat loss. The skin humidity reduction follows $\delta q_{\mathrm{skin}} = C_{\mathrm{wet}}\delta T_{\mathrm{skin}}$ and feeds back on latent heat loss in the full iteration.

Solar heating is a separate influence. Daytime warming below the skin can coexist with interface cooling. A cool-skin correction alone is not a warm-layer model. Observations and updated COARE work find larger cool-skin differences in light winds; see Fairall et al. (2026). The figure here uses the ROMS skin equations, not that updated formulation.

10. Turbulent heat, rain and the Webb correction

The returned sensible and latent losses include corrections beyond the basic similarity products.

The leading turbulent terms are

$$H_s = -c_{pa}\rho_a u_* T_*, \qquad H_l = -L_v \rho_a u_* q_*. \tag{37}$$

Rain sensible heat. Let $P =$ `rain` in kg m^{-2} s^{-1}. Define

$$D_v = 2.11 \times 10^{-5}(T_{aK}/273.16)^{1.94}, \tag{38}$$

$$D_h = \frac{0.02411[1 + T_{aC}(3.309 \times 10^{-3} - 1.44 \times 10^{-6} T_{aC})]}{\rho_a c_{pa} + \varepsilon}, \quad A_q = \frac{q_{a,\mathrm{sat}} L_v}{R_d T_{aK}^2}, \tag{39}$$

$$f_{\mathrm{wet}} = \left[1 + 0.622 \frac{A_q L_v D_v}{c_{pa} D_h}\right]^{-1}, \qquad H_{\mathrm{rain}} = |P| f_{\mathrm{wet}} c_{pw} \left[(T_s - T_a) + \frac{L_v}{c_{pa}}(q_s - q_a)\right]. \tag{40}$$

The wet-bulb factor represents coupled heat and vapor diffusion around falling drops. Notice that this expression uses the bulk contrasts, not the skin-corrected contrasts. It can change sign with the thermodynamic state.

Webb contribution. Heat and moisture exchange affect air density and its mean vertical mass transport. The code accounts for this in latent heat with

$$w_{\mathrm{Webb}} = -1.61 u_* q_* - (1 + 1.61 q_a)\frac{u_* T_*}{T_{aK}}, \qquad H_{\mathrm{Webb}} = \rho_a L_v q_a w_{\mathrm{Webb}}. \tag{41}$$

This velocity is neither ocean vertical velocity nor gustiness.

```
Hs=-blk_Cpa*rhoAir(i)*Wstar(i)*Tstar(i)
SHeat(i,j)=Hs+Hsr
Hl=-Hlv(i,j)*rhoAir(i)*Wstar(i)*Qstar(i)
upvel=-1.61_r8*Wstar(i)*Qstar(i)                          &
 & -(1.0_r8+1.61_r8*Q(i))*Wstar(i)*Tstar(i)/TairK(i)
Hlw=rhoAir(i)*Hlv(i,j)*upvel*Q(i)
LHeat(i,j)=Hl+Hlw
```

$$\boxed{H_{\mathrm{sens}} = H_s + H_{\mathrm{rain}}, \qquad H_{\mathrm{lat}} = H_l + H_{\mathrm{Webb}}.} \tag{42}$$

Both are positive out of the ocean under ocean-cooling conditions. In the `REGCM_COUPLING` branch the assignments to `SHeat` and `LHeat` are skipped, so the ordinary open-water output formula must not be assumed to replace the coupled flux inputs.

11. Radiation: three longwave paths

Longwave and shortwave enter the final heat budget with a positive-downward convention.

At input, radiative flux arrays are in kinematic temperature-flux units. The routine temporarily recovers W/m^2 using

$$H_{\text{scale}} = \rho_0 c_{p,o}, \qquad Q_{SW} = H_{\text{scale}}\,\texttt{srflx}. \tag{43}$$

$c_{p,o}$ is the ROMS ocean heat capacity `Cp`, distinct from air c_{pa} and the bulk-parameterization water value c_{pw}. Let $\mathcal{E} = 0.97$ be infrared emissivity and $\sigma = 5.67 \times 10^{-8}$ W m^{-2} K^{-4}.

A. `LONGWAVE`: empirical net longwave. The Berliand branch uses a separate saturation-pressure expression,

$$e_{s,LW} = 10^{(0.7859+0.03477T_{aC})/(1+0.00412T_{aC})}, \qquad e_{a,LW} = \mathrm{RH}e_{s,LW}, \tag{44}$$

$$\boxed{Q_{LW} = -\mathcal{E}\sigma\left[T_{aK}^4(0.39 - 0.05\sqrt{e_{a,LW}})(1 - 0.6823C^2) + 4T_{aK}^3(T_{sK} - T_{aK})\right].} \tag{45}$$

C is cloud fraction. Vapor pressure is in hPa, as required by these empirical coefficients. Moisture and clouds reduce radiative cooling by changing effective atmospheric emission.

B. `LONGWAVE_OUT`: prescribed incoming longwave.

$$\boxed{Q_{LW} = Q_{LW,\downarrow}^{\text{input}} - \mathcal{E}\sigma T_{sK}^4.} \tag{46}$$

The code subtracts outgoing sea-surface emission from the supplied incoming flux. This is the literal expression; it does not separately multiply the incoming value by $\mathcal{E}$ in this block. A coupled input may be restored from the last exchange before calculation.

C. Neither flag: prescribed net longwave.

$$\boxed{Q_{LW} = H_{\text{scale}}\,\texttt{lrflx}^{\text{input}}.} \tag{47}$$

A negative value is a net ocean heat loss. These branches are alternatives, with `LONGWAVE` taking precedence.

Humidity compatibility caveat. The empirical `LONGWAVE` branch multiplies its saturation pressure by `RH=Hair` before the later humidity-input branch is evaluated. It therefore expects fractional RH here, even though the subsequent bulk humidity conversion can accept specific humidity in g/kg. Those two input interpretations are not interchangeable in this option.

12. Final stress vectors: use the relative-wind direction

Gustiness affects the calculated stress strength, but the resolved relative wind sets its direction.

Rain contributes a scalar momentum flux

$$\tau_{\text{rain}} = 0.85|P|U_{\text{rel,mag}}. \tag{48}$$

The source then combines it with turbulent stress:

```
Taur=0.85_r8*ABS(rain(i,j))*Wmag(i)
cff=rhoAir(i)*(Wstar(i)*Wstar(i)+Taur/rhoAir(i))        &
 & /(Wmag(i)+eps)
Taux(i,j)=cff*Uair(i,j)
Tauy(i,j)=cff*Vair(i,j)
```

$$\boxed{\tau_x = \frac{\rho_a {u_*}^2 + \tau_{\text{rain}}}{U_{\text{rel,mag}} + \varepsilon} U_{\text{rel}}, \qquad \tau_y = \frac{\rho_a {u_*}^2 + \tau_{\text{rain}}}{U_{\text{rel,mag}} + \varepsilon} V_{\text{rel}}.} \tag{49}$$

The physical stresses are in Pa (N/m^2), directed along the relative wind. For $U_{\text{rel,mag}} \gg \varepsilon$, the magnitude is approximately $\rho_a {u_*}^2 + \tau_{\text{rain}}$.

Why not divide by U_{eff}? Since $\sqrt{U_{\text{rel}}^2 + V_{\text{rel}}^2} = U_{\text{rel,mag}}$, dividing by the gustiness-enhanced speed would multiply the intended vector magnitude by $U_{\text{rel,mag}}/U_{\text{eff}} < 1$. The source records this as a February 2024 correction by Wilkin and Pareja-Roman. `Wspeed` is recomputed after the loop, but it is not the open-water vector-normalization denominator.

Exactly calm resolved relative wind. If $U_{\text{rel}} = V_{\text{rel}} = 0$, both stress components are zero in this implementation even if gustiness keeps scalar heat/moisture exchange nonzero. A nonzero scalar u_* does not define a resolved stress direction in that limit.

ROMS receives kinematic stress on its staggered grid:

$$\boxed{\texttt{sustr}_{i,j} = \frac{\tau_{x,i-1,j} + \tau_{x,i,j}}{2\rho_0}, \qquad \texttt{svstr}_{i,j} = \frac{\tau_{y,i,j-1} + \tau_{y,i,j}}{2\rho_0}.} \tag{50}$$

These arrays have units m^2/s^2, not Pa. Land and wet/dry masks are applied when enabled; periodic and MPI exchanges follow. On a curvilinear grid these are grid-axis components, which must be rotated back if east/north stress vectors are required.

13. Final heat and freshwater outputs

Positive net heat flux warms the ocean. Positive evaporation removes freshwater.

For the ordinary open-water path, Eq. (51) gives the dimensional net heat flux:

$$\boxed{Q_{\text{net,down}} = Q_{SW} + Q_{LW} - H_{\text{sens}} - H_{\text{lat}}.} \tag{51}$$

The subtraction converts the upward-positive turbulent losses to the ocean's downward-positive heating convention.

```
Hscale=1.0_r8/(rho0*Cp)
lrflx(i,j)= LRad(i,j)*Hscale
lhflx(i,j)=-LHeat(i,j)*Hscale
shflx(i,j)=-SHeat(i,j)*Hscale
stflux(i,j,itemp)=srflx(i,j)+lrflx(i,j)                 &
 &              +lhflx(i,j)+shflx(i,j)
```

$$\boxed{\texttt{stflux(temperature)} = \frac{Q_{\text{net,down}}}{\rho_0 c_{p,o}}} \quad [^\circ\text{C m/s}]. \tag{52}$$

The individual output arrays `srflx,lrflx,lhflx,shflx` also use these kinematic units within this routine. Recover W/m^2 by multiplying by $\rho_0 c_{p,o}$; do not apply that conversion twice when reading output that has already been scaled by an I/O convention.

With `EMINUSP`, the mass evaporation rate is

$$\boxed{E = \frac{H_{\text{lat}}}{L_v + \varepsilon}} \quad [\text{kg m}^{-2}\text{s}^{-1}], \qquad \boxed{F_{E-P} = \frac{E - P}{\rho_{fw}}} \quad [\text{m/s}]. \tag{53}$$

The denominator uses freshwater density `rhow`. Positive $E - P$ removes freshwater and tends to increase ocean salinity. At this stage `stflux(isalt)` stores F_{E-P}; the source comments specify multiplication by surface salinity later in `set_vbc.F`. It is not yet a salinity-times-velocity flux here.

Output	Physical interpretation	Internal units
`sustr,svstr`	Stress divided by ρ_0	m^2/s^2
`lhflx,shflx`	Downward-positive latent/sensible	°C m/s
`lrflx,srflx`	Downward-positive radiation	°C m/s
`stflux(itemp)`	Net ocean heating	°C m/s
`evap`	Upward-positive water mass loss	kg m^{-2} s^{-1}
`stflux(isalt)`	$E - P$ before salinity factor	m/s

Appendix A. The stability functions actually used

For unstable conditions $\zeta < 0$, the code blends Kansas and free-convection forms using

$$f_w = \frac{\zeta^2}{1+\zeta^2}, \qquad \psi = (1-f_w)\psi_K + f_w\psi_C. \tag{54}$$

Momentum:

$$x_m = (1-15\zeta)^{1/4}, \quad \psi_{mK} = 2\ln\frac{1+x_m}{2} + \ln\frac{1+x_m^2}{2} - 2\tan^{-1}x_m + \frac{\pi}{2}, \tag{55}$$

$$y_m = (1-10.15\zeta)^{1/3}, \quad \psi_{mC} = \frac{3}{2}\ln\frac{1+y_m+y_m^2}{3} - \sqrt{3}\tan^{-1}\frac{1+2y_m}{\sqrt{3}} + \frac{\pi}{\sqrt{3}}. \tag{56}$$

Heat and moisture:

$$x_h = (1-15\zeta)^{1/2}, \qquad \psi_{hK} = 2\ln\frac{1+x_h}{2}, \tag{57}$$

$$y_h = (1-34.15\zeta)^{1/3}, \quad \psi_{hC} = \frac{3}{2}\ln\frac{1+y_h+y_h^2}{3} - \sqrt{3}\tan^{-1}\frac{1+2y_h}{\sqrt{3}} + \frac{\pi}{\sqrt{3}}. \tag{58}$$

For stable conditions $\zeta \geq 0$, let $c = \min(50, 0.35\zeta)$. Then

$$\boxed{\psi_m = -\left[(1+\zeta) + 0.6667(\zeta - 14.28)e^{-c} + 8.525\right]}, \tag{59}$$

$$\boxed{\psi_h = -\left[(1+2\zeta)^{3/2} + 0.6667(\zeta - 14.28)e^{-c} + 8.525\right]}. \tag{60}$$

The argument is evaluated at the corresponding measurement height. The exponential cap limits extreme stable-function evaluation. Rounded coefficients imply the stable expression is only approximately zero at exact neutral conditions.

Physical reading of the profile denominators

Unstable stratification generally promotes turbulent transport and reduces the resistance represented by $\ln(z/z_0) - \psi$. Stable stratification suppresses transport and increases that resistance. These functions are empirical surface-layer closures, not an explicit simulation of all atmospheric turbulent motions.

Scope and source notes

This handout follows the inspected source, including details that deserve verification.

1. **Three fixed passes.** No convergence tolerance is computed. Final gustiness/skin values are one update ahead of the stored scales.
2. **Humidity input is branch-dependent.** Fractional RH and specific humidity in g/kg share a threshold heuristic; empirical longwave separately expects fractional RH.
3. **Literal gustiness denominator and cool-skin longwave sign are flagged.** Sections 8–9 reproduce the code rather than silently substituting a textbook or reference-COARE expression. These are review points, not changes made to the model.
4. **The explanation does not establish enabled build options.** COARE 3.5 is the default branch when `COARE_30` is absent. Wave roughness, drag limits, cool skin, radiation, current subtraction and freshwater outputs remain conditional.
5. **Sea-ice paths are separate.** The file also computes ice stress, ice/snow heat terms, precipitation phase corrections, ice-cover radiation and ice thermodynamic coefficients. They are not represented by the open-water final-budget equations above. In particular `Qnet_ao` in the ice branch reverses the sign of the kinematic temperature flux to report outward heat in W/m^2.

Primary source. ROMS source file `ROMS/Nonlinear/bulk_flux.F`; ROMS version **4.3**; repository revision `31cabd31`. The inspected source has SHA-256 prefix `b5fa33b42d0fb853`. Constants and measurement-height defaults were checked in `ROMS/Modules/mod_scalars.F`; wind rotation was checked in `ROMS/Nonlinear/set_data.F`. The repository identifier alone does not assert that the working tree is pristine.

Background reading. The file cites Fairall et al. (1996, 2003), Liu et al. (1979), and Edson et al. (2013), among others. NOAA's COARE algorithm repository and Ocean Climate Stations flux documentation provide reference context. The equations in this handout are keyed to the inspected ROMS implementation; it should not be assumed identical in every detail to a separately distributed COARE release.

Final reference: what the ocean receives

Three scales, one relative-wind direction, and an explicit heat-flux sign convention.

$$u_* \text{ friction velocity}, \qquad T_* \text{ temperature scale}, \qquad q_* \text{ humidity scale}. \tag{61}$$

Relative wind and effective exchange speed

$$U_{\mathrm{rel}} = U_{\mathrm{wind}} - u_{\mathrm{oce}}, \qquad V_{\mathrm{rel}} = V_{\mathrm{wind}} - v_{\mathrm{oce}}, \tag{62}$$

$$U_{\mathrm{rel,mag}} = \sqrt{U_{\mathrm{rel}}^2 + V_{\mathrm{rel}}^2}, \qquad U_{\mathrm{eff}} = \sqrt{U_{\mathrm{rel,mag}}^2 + u_{\mathrm{gust}}^2}. \tag{63}$$

Current subtraction applies only when its flag is enabled. All components must share the same coordinate basis.

Physical stress vector, positive along the relative wind

$$\boxed{\begin{pmatrix}\tau_x \\ \tau_y\end{pmatrix} = \frac{\rho_a {u_*}^2 + 0.85|P|U_{\mathrm{rel,mag}}}{U_{\mathrm{rel,mag}} + \varepsilon}\begin{pmatrix}U_{\mathrm{rel}} \\ V_{\mathrm{rel}}\end{pmatrix}} \qquad [\mathrm{Pa}]. \tag{64}$$

ROMS averages these components to their staggered velocity points and divides by ρ_0 to obtain `sustr` and `svstr` in $\mathrm{m}^2/\mathrm{s}^2$.

Sensible and latent heat losses, positive upward

$$\boxed{H_{\mathrm{sens}} = -\rho_a c_{pa} u_* T_* + H_{\mathrm{rain}}}, \tag{65}$$

$$\boxed{H_{\mathrm{lat}} = -\rho_a L_v u_* q_* + H_{\mathrm{Webb}}} \qquad [\mathrm{W/m^2}]. \tag{66}$$

Rain and Webb corrections are defined in Section 10. Radiation is defined in Section 11.

Net ocean heating, positive downward

$$\boxed{Q_{\mathrm{net,down}} = Q_{SW} + Q_{LW} - H_{\mathrm{sens}} - H_{\mathrm{lat}}}, \qquad \boxed{\texttt{stflux(itemp)} = \frac{Q_{\mathrm{net,down}}}{\rho_0 c_{p,o}}}. \tag{67}$$

A positive net value warms the ocean. These summaries apply to the ordinary open-water, non-RegCM path before masking. The stated sea-ice and coupled-flux exceptions remain conditional.